



Researchers are developing new display sensors for smartphones that will read the user's split to not only detect the body temperature, but also analyze the DNA (Courtesy: <http://www.indiatimes.com>)

The researchers have also developed first-of-its-kind laser-written guiding system. It can further help these advancements and use laser into the glass to create pathways which transmit data in form of small light beams, as waveguides act as a tunnel to channel these lights. It is similar to how wires convey electric signals. If it is added to a smartphone along with readable code, it can open the doors to many opportunities by such device makers.

## Images from your mind on a screen

Researchers in the US claim to have built a unique machine that can extract images from people's brains and then project them on a display screen. Yes, you heard that right! Alan Cowen, a neuroscientist at the University of California, Berkeley and his co-researchers, Brice Kuhl of New York University and Professor Marvin Chun of Yale, have built this one-of-its-kind system that envisions a future where cops and investigative agencies could generate images of criminals from the mind of a witness!

The machine has been used to accurately reconstruct human faces based on data received from a brain scanner, until now. However, the same approach could be used to extract images from people's dreams, memories and imagination, in the coming days. The methods yield strikingly accurate neural reconstructions of faces.

For research purposes, six volunteers were shown as many as 300 faces

while they lay in an MRI scanner. The process showed how their brains would respond to different anatomical features they saw. This gave the researchers a database of responses, across several areas of the brain. The volunteers were then shown a set of entirely new faces and their responses to the new images were measured. The researchers compared these responses to the database, following which they were able to reconstruct the image the volunteers were looking at.

## Auto-follow sports drone set for take-off

Stratasys, a leading global provider of 3D printing and additive manufactur-



3D-printed auto-follow drone (Courtesy: <http://www.unmanned-aerial.com>)

ing solutions, has announced that UAV start-up company, Helico, has successfully introduced AirDog, the world's first automated drone designed to track and video outdoor sports and activities, using Stratasys 3D printing technology. Aimed primarily at the consumer market, AirDog is an innovative, yet simple-to-use 'quad-copter' that operates via a wrist-worn tracking device and accommodates a standard GoPro sports camera.

Users can automatically capture exciting live aerial video footage and still photography of themselves, having set distance, speed and height levels for AirDog to follow. Helico is specifically targeting the outdoor 'extreme' sports

market and expects AirDog to be of particular interest to recreational participants of freestyle BMX, motocross and skateboarding, as well as water-sports such as surfing, kite-surfing and wake-boarding.

## Intellectual property for capacitive-touch sensing devices

Renesas Electronics has developed an innovative new intellectual property (IP) that implements industry-leading capacitive-touch sensing technology suitable for home appliances and healthcare equipment, such as blood pressure and blood glucose meters. The IP achieves high-touch sensitivity five times better than of Renesas'

R8C/3xT microcontrollers, as well as high noise immunity allowing the technology to pass strict noise tests. Renesas IP supports the mutual capacitance method which is more reliable and versatile than the common self-capacitance method.

Renesas will also provide development tools to make application development easier. For human machine interface applications for electric and electronic equipment, touch-key interfaces have been the subject of increasing attention. The use of capacitive-touch technology is being rapidly adopted because it can easily improve reliability in product design and enhance the end-user experience.

For home appliances used in the kitchen, it is often the case that the appliance will be splashed with water, or the user will be wearing gloves. Besides, equipment such as IH stoves emit strong electromagnetic fields. So a technology that provides both high sensitivity and high noise immunity can reliably respond to the static capacitance when such end products are used.

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